Validation Execution Instantiation & Control Module (VEICM)

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Governed Space

Validation Execution Instantiation & Control

Canonical Definition

Validation Execution Instantiation & Control Module (VEICM) defines the governance framework for transforming an authorized Validation Execution Basis into an active, bounded, controlled, and identifiable Validation Execution Instance operating against the approved Validation Object under the required methodological, operational, environmental, evidentiary, source, participant, system, and sequencing conditions.

It establishes the conditions under which authorized validation moves from **execution readiness** into **controlled operational performance.**

Operational Role

VEICM serves as the second internal governance space of the Validation Execution Standard.

The preceding module establishes:

VEBAM — Is there a sufficiently complete, coherent, execution-ready, versioned, and authorized basis from which validation may begin?

VEICM asks:

How is that authorized basis instantiated into actual validation activity while preserving the controls, boundaries, conditions, roles, sequence, evidence relationships, and methodological requirements governing execution?

The governed progression is:

Authorized Validation Execution Basis → Execution Activation → Object Binding → Method Instantiation → Role Activation → Environment Establishment → Control Activation → Sequenced Validation Activity → Controlled Validation Execution Instance

VEICM does not determine what the execution ultimately demonstrates.

It governs how the approved validation structure becomes operational reality.

Module Operational Space

VEICM governs:

- execution instantiation,
- execution activation,
- Validation Object binding,
- object-version binding,
- method instantiation,
- criteria activation,
- evidence activation,
- source-condition activation,
- execution-role activation,
- participant-role assignment,
- system activation,
- instrument activation,
- environment establishment,
- execution boundaries,
- execution controls,
- control verification,
- execution sequencing,
- activity ordering,
- execution dependencies,
- execution synchronization,
- human execution,
- automated execution,
- hybrid execution,
- execution handoffs,
- intervention control,
- override control,
- access control,
- configuration control,
- execution-state control,
- execution pause,
- execution resume,

- execution termination,
- and execution-control traceability.

Module Function

VEICM applies after a Validation Execution Basis has been sufficiently established and authorized.

Its function is to prevent:

- authorization being mistaken for actual execution,
- validation beginning against the wrong object instance,
- method documentation being mistaken for method instantiation,
- execution occurring outside approved boundaries,
- participants performing undefined roles,
- systems operating under incorrect configurations,
- instruments being used outside required conditions,
- required controls remaining inactive,
- criteria being disconnected from execution activities,
- evidence being used outside its approved role,
- Source Reliability restrictions being ignored,
- required sequencing being lost,
- uncontrolled human intervention,
- undocumented automated behavior,
- ungoverned human-machine handoffs,
- and active execution drifting away from the authorized Validation Execution Basis.

Instantiation

Instantiation is the process through which a governed validation design becomes a specific operational validation instance.

VALIDOS® distinguishes:

Method Definition

from:

Execution Instantiation

A method may describe a reusable procedure.

Instantiation binds that procedure to:

- one Validation Object,
- one object version or state,
- one Execution Basis,
- applicable Validation Criteria,
- defined evidence,
- defined sources,
- specific participants,

- specific systems,
- specific instruments,
- a specific environment,
- and a specific execution period.

Without this binding, a method remains a design rather than an executed validation process.

Execution Activation

Execution Activation marks the transition from:

Authorized but Not Yet Executing

to:

Active Validation Execution

Activation should occur only when required start conditions remain satisfied.

Material activation information may include:

- Execution Instance,
- activation time,
- active Validation Object,
- active Basis version,
- active Method version,
- execution environment,
- participants,
- systems,
- instruments,
- and applicable controls.

Activation Gate

VEICM may establish an **Activation Gate** immediately before operational execution begins.

The gate may verify that:

- authorization remains valid,
- the Basis Freeze remains applicable,
- the Validation Object has not materially changed,
- required resources remain available,
- systems and instruments remain ready,
- required evidence remains available,
- source conditions remain current,
- required participants are present,
- and critical execution controls can be activated.

A material failure at the Activation Gate may prevent execution from beginning.

Validation Object Binding

Every Validation Execution Instance must remain bound to the Validation Object it is authorized to examine.

Object binding may include:

- object identity,
- object version,
- configuration,
- state,
- environment,
- temporal reference,
- or another material object condition.

This establishes:

Execution Instance X → Validation Object Y → Object State Z

Object Verification at Activation

The Validation Object should be verified at or immediately before execution activation where object substitution or change is materially possible.

Verification may establish:

- correct identity,
- correct version,
- correct configuration,
- correct physical instance,
- correct digital instance,
- correct dataset,
- correct model,
- correct environment,
- or another relevant condition.

Object-State Continuity

Some Validation Objects change while validation is being executed.

Examples include:

- adaptive AI systems,
- continuously updated datasets,
- autonomous systems,

- operational infrastructure,
- live software systems,
- or dynamic physical processes.

VEICM governs whether the object state must be:

- frozen,
- monitored,
- versioned,
- periodically captured,
- or otherwise controlled during execution.

Method Instantiation

Method Instantiation transforms the approved Validation Method into specific executable activities.

This may require establishing:

- procedure sequence,
- test configuration,
- sampling parameters,
- analytical settings,
- simulation settings,
- repetition count,
- thresholds,
- observation periods,
- comparison conditions,
- or other method-specific parameters.

The instantiated method should remain traceable to the approved method.

Method Parameterization

A Validation Method may contain parameters that must be set for a specific execution.

Examples include:

- sample size,
- test duration,
- model temperature,
- simulation seed,
- measurement frequency,
- tolerance,
- operating load,
- confidence threshold,
- or another execution parameter.

VEICM governs whether those parameters remain within the approved methodological boundaries.

Criteria Activation

Applicable Validation Criteria become active execution requirements.

Criteria activation establishes which validation activities are expected to produce information relevant to each criterion.

Conceptually:

Criterion → Required Activity → Execution Output

This does not determine criterion satisfaction.

It creates operational connectivity between criteria and execution.

Evidence Activation

Qualified Validation Evidence may become active inputs to execution.

VEICM preserves:

- evidence identity,
- evidence version,
- Evidence Fitness status,
- evidence limitations,
- applicability conditions,
- temporal conditions,
- and permitted evidentiary role.

Evidence should not silently acquire a broader role during execution than the role for which it was qualified.

Source-Condition Activation

Source Reliability conditions follow evidence into execution.

If a source was determined:

Conditionally Reliable the relevant conditions become execution controls where applicable.

Examples include:

- independent corroboration required,
- use restricted to one role,
- defined version only,
- limited population,

- limited measurement range,
- or specific oversight requirement.

Execution Role Activation

Execution may involve multiple roles.

These may include:

- Validation Executor,
- Validation Observer,
- Instrument Operator,
- Data Provider,
- Expert Reviewer,
- System Operator,
- AI Executor,
- Autonomous Agent,
- Supervisor,
- or another defined role.

VEICM activates the roles required for the specific
Execution Instance.

Role Assignment

A participant should not automatically gain authority to perform
every execution activity merely because they participate
in validation.

Role assignment may establish:

- permitted activities,
- restricted activities,
- intervention rights,
- approval rights,
- observation rights,
- override rights,
- and escalation responsibilities.

Role Separation

Some validation contexts may require separation between roles.

For example:

System Operator ≠ Independent Validator

or:

Evidence Provider ≠ Final Execution Controller

where independence or accountability requires separation.

VEICM does not impose universal separation.

It preserves required separation where established by the Validation Execution Basis.

Human Execution Control

Where humans perform validation activities, VEICM may govern:

- task assignment,
- procedural instructions,
- execution order,
- required confirmations,
- access permissions,
- manual interventions,
- role boundaries,
- and escalation conditions.

Human judgment may remain necessary.

The governance requirement is that material human execution remains attributable and bounded.

Automated Execution Control

Where automated systems perform validation activities, VEICM may govern:

- system identity,
- software version,
- configuration,
- execution permissions,
- parameter settings,
- input sources,
- tool access,
- output destination,
- failure handling,
- and operational boundaries.

Automation must not transform execution into an unobservable methodological black box.

AI Execution Control

Where an AI system performs material validation activities, execution control may additionally govern:

- model identity,
- model version,
- system instructions,
- tool permissions,
- context inputs,
- memory conditions,
- retrieval sources,
- sampling configuration,
- execution limits,
- human oversight,
- and escalation conditions.

VEICM does not determine whether the AI output is correct.

It establishes the governed conditions under which the AI performed the validation activity.

Autonomous Execution Control

Where autonomous systems execute validation activities, VEICM may govern:

- delegated authority,
- execution boundaries,
- permitted actions,
- system state,
- environmental access,
- escalation,
- human override,
- and termination conditions.

Autonomous execution must remain connected to the authorized Execution Basis.

Hybrid Execution

Validation may combine:

Human → Machine

Machine → Human

Machine → Machine

or:

Human → Machine → Human

execution relationships.

VEICM governs the material handoffs between these execution actors.

Execution Handoff

An Execution Handoff occurs when responsibility for a validation activity or execution state passes between participants or systems.

A material handoff may identify:

- originating actor,
- receiving actor,
- transferred task,
- transferred evidence,
- transferred execution state,
- time,
- and relevant conditions.

This prevents responsibility from disappearing between execution stages.

Environment Establishment

The validation environment must correspond sufficiently to the conditions required by the Validation Execution Basis.

The environment may include:

- physical location,
- digital infrastructure,
- network state,
- system dependencies,
- simulation environment,
- testing sandbox,
- production environment,
- laboratory,
- operational facility,
- or another relevant context.

Environment Verification

Where environmental conditions materially affect validation, VEICM may verify them before or during execution.

Examples include:

- temperature,
- humidity,
- network latency,
- system load,
- hardware configuration,

- software dependencies,
- access restrictions,
- simulation configuration,
- or another material condition.

Control Activation

Required execution controls should become active when execution begins.

Controls may include:

- access controls,
- configuration locks,
- audit logging,
- environmental controls,
- sequencing controls,
- evidence controls,
- source controls,
- safety controls,
- human oversight,
- automated monitoring,
- or termination controls.

Control Verification

A control that is defined but inactive provides no operational protection.

VEICM therefore distinguishes:

Control Required

from:

Control Active

and:

Control Verified

Where material, control activation should be demonstrable.

Execution Boundary

An Execution Boundary defines the operational limits within which validation activity may occur.

Boundaries may concern:

- Validation Object,
- system access,
- data access,
- permitted tools,
- physical environment,
- time,
- actions,
- scope,
- participant authority,
- or another relevant condition.

Execution outside a material boundary becomes a governance event.

Configuration Control

Validation execution may depend upon precise configuration.

Configuration control may apply to:

- Validation Object,
- instruments,
- software,
- models,
- datasets,
- testing systems,
- simulation environments,
- or supporting infrastructure.

Material configuration change should remain detectable.

Access Control

Execution participants and systems may require access to:

- evidence,
- systems,
- instruments,
- datasets,
- operational environments,
- or Validation Object components.

VEICM governs whether access corresponds to assigned execution roles.

Excessive access may create integrity or independence concerns.

Insufficient access may prevent legitimate execution.

Execution Sequencing

Where order affects validation validity, VEICM governs the required
execution sequence.

A sequence may include:

Initialize → Observe → Measure → Test → Compare → Record → Confirm

or another method-specific structure.

Sequence governance ensures that activities do not occur in an order
that changes their methodological meaning.

Execution Dependency

Some validation activities depend upon successful completion
of others.

For example:

Activity B may require Activity A to complete successfully.

VEICM makes these dependencies explicit where material.

Conditional Activity

An execution activity may occur only if a defined condition
is satisfied.

For example:

If Criterion A produces Condition X → Execute Test B

Conditional activities should remain governed rather than improvised
during execution.

Parallel Execution

Some validation activities may occur simultaneously.

Parallel execution may require:

- synchronization,
- state consistency,
- shared-resource control,
- time alignment,
- or output coordination.

VEICM governs these conditions where parallel activity affects
methodological integrity.

Execution Synchronization

Where multiple systems, instruments, or participants contribute to the same validation activity, their execution states may require synchronization.

Examples include:

- synchronized timestamps,
- coordinated measurement windows,
- shared object state,
- aligned simulation state,
- or common configuration.

Unsynchronized activity may create misleading evidence.

Execution State

VEICM may distinguish operational execution states such as:

Not Activated

Active

Paused

Resuming

Terminating Completed

or:

Aborted

These states describe operational execution.

They do not represent Validation State.

Execution Pause

Execution may be intentionally paused where:

- a prerequisite changes,
- a safety condition occurs,
- an instrument requires review,
- an unexpected object change appears,
- required evidence becomes unavailable,
- or another material condition requires assessment.

A pause preserves the execution instance while stopping further activity.

Execution Resume

Before paused execution resumes, VEICM may require confirmation that:

- authorization remains valid,
- object state remains acceptable,
- required controls remain active,
- environmental conditions remain appropriate,
- and the interruption has not invalidated prior activity.

Execution Termination

Execution may be terminated where continuation is no longer legitimate or useful.

Termination may result from:

- critical control failure,
- invalid object state,
- authorization withdrawal,
- safety condition,
- unrecoverable system failure,
- or another material event.

Termination does not automatically determine the Validation Object's validity.

Manual Intervention

Manual intervention occurs when a human changes, redirects, corrects, pauses, overrides, or otherwise materially affects automated or predefined execution.

Intervention may be legitimate.

Material intervention should remain identifiable.

Override Control

Where override authority exists, VEICM may govern:

- who may override,
- what may be overridden,
- under which conditions,
- for how long,
- and what record is required.

An override should not silently rewrite the execution history.

Execution Drift

Execution Drift occurs when active validation gradually moves away from the authorized Execution Basis without one clearly identifiable deviation event.

Examples include:

- accumulating parameter changes,
- expanding scope,
- evolving system configuration,
- repeated informal workarounds,
- changing evidence sources,
- or shifting participant roles.

VEICM seeks to detect and constrain execution drift before it undermines validation integrity.

Control Loss

A **Control Loss** occurs where a required execution control becomes unavailable, ineffective, bypassed, or materially degraded.

Control Loss may require:

- pause,
- correction,
- escalation,
- deviation classification,
- partial repetition,
- or termination.

The response depends upon materiality.

Control Recovery

Where a lost control is restored, VEICM preserves:

- when control was lost,
- which execution activities occurred during the loss,
- when it was restored,
- and whether affected activities remain usable.

Restoration does not automatically validate activity performed while control was absent.

Instantiation Conformity

VEICM may assess whether the operational Execution Instance sufficiently corresponds to the authorized Validation Execution Basis.

Possible conditions may include: **Instantiation Conformant**

Instantiation Conditionally Conformant

Instantiation Deviation Identified

Instantiation Non-Conformant

or:

Instantiation Undetermined

This is an execution-control assessment.

It is not final Execution Conformity under VEXS.

Control Continuity

Controls should remain active for as long as their governed purpose requires.

VEICM therefore governs not only control activation but also control continuity across execution.

A control active only at the beginning of validation may not satisfy a requirement for continuous control.

Minimum Implementation Framework

1. Activate the Authorized Execution Instance

Confirm that the approved Validation Execution Basis remains applicable and activate the execution instance.

2. Bind the Validation Object

Connect execution to the exact object, version, configuration, and state being validated.

3. Instantiate the Validation Method

Transform the approved method into concrete executable activities and parameters.

4. Activate Criteria, Evidence, and Source Conditions

Connect applicable criteria, qualified evidence, and Source Reliability conditions to operational execution.

5. Activate Execution Roles

Assign the human, automated, AI, autonomous, or hybrid participants required for execution.

6. Establish the Execution Environment

Verify required operational, physical, digital, or simulation conditions.

7. Activate and Verify Controls

Ensure required controls are operational before dependent validation activity proceeds.

8. Govern Sequence and Dependencies

Preserve required activity order, conditional activities, synchronization, and dependencies.

9. Govern Interventions and Handoffs

Maintain visibility over manual intervention, overrides, and responsibility transfers.

10. Govern Execution State

Control activation, pause, resume, termination, and completion states.

11. Preserve Control Traceability

Maintain:

- Execution Instance,
- active Basis version,
- Validation Object binding,
- object state,
- Method instance,
- method parameters,
- Criteria activation,
- Evidence activation,
- Source Reliability conditions,
- participant roles,
- system identities,
- instrument identities,
- environment,
- controls,
- access,
- configuration,
- sequence,
- dependencies,
- handoffs,

- interventions,
- overrides,
- execution state,
- control loss,
- control recovery,
- and sufficient lifecycle traceability.

Governance Outputs

VEICM may produce:

- Validation Execution Activation Record,
- Execution Activation Gate Record,
- Validation Object Binding Record,
- Object-State Verification Record,
- Validation Method Instance,
- Method Parameterization Record,
- Validation Criteria Activation Map,
- Validation Evidence Activation Record,
- Source Condition Activation Record,
- Execution Role Register,
- Participant Role Assignment,
- System Activation Record,
- Instrument Activation Record,
- Execution Environment Record,
- Environment Verification Record,
- Execution Boundary Record,
- Execution Control Register,
- Control Activation Record,
- Control Verification Record,
- Configuration Control Record,
- Access Control Record,
- Execution Sequence Map,
- Execution Dependency Map,
- Execution Synchronization Record,
- Execution Handoff Record,
- Manual Intervention Record,
- Override Record,
- Execution State Record,
- Execution Pause Record,
- Execution Resume Record,
- Execution Termination Record,
- Control Loss Record,
- Control Recovery Record,
- Execution Drift Record,
- Instantiation Conformity Assessment,
- and Execution Control Traceability Record.

Use Case 1 — AI Model Validation

Scenario

An AI model has an authorized Validation Execution Basis covering a defined model version, benchmark set, test methodology, and execution environment.

Application

VEICM binds execution to the exact model version, activates the approved benchmark evidence, instantiates the required test parameters, records the model configuration, activates logging, assigns human oversight, and establishes the required test sequence.

During execution, an operator attempts to change one model parameter.

Result

The parameter change becomes a governed intervention rather than an invisible modification of the test environment.

The execution history preserves the exact conditions under which each validation activity occurred.

Use Case 2 — Industrial Validation

Scenario

Several sensors and two operators must perform synchronized measurements on an industrial system.

Application

VEICM verifies instrument identity, activates measurement controls, binds operators to assigned roles, synchronizes measurement windows, establishes the required sequence, and records the active system configuration.

Result

The resulting measurements remain connected to the actual operational conditions under which they were produced.

Use Case 3 — Autonomous Validation Agent

Scenario An autonomous AI agent performs a large set of validation tests across a software environment.

Application

VEICM establishes:

- agent identity,
- execution permissions,
- accessible tools,
- system boundaries,

- test sequence,
- permitted actions,
- logging requirements,
- escalation conditions,
- and human override authority.

Result

The autonomous agent may execute validation activity at scale
without becoming methodologically unbounded.

Its operational actions remain connected to the authorized
Validation Execution Basis.

Architectural Position

Validation Execution Instantiation & Control is the second internal
governance space of the **Validation Execution Standard (VEXS).**

VEBAM establishes:

What governed basis authorizes validation execution to begin?

VEICM establishes:

How does that basis become a controlled operational Validation Execution Instance?

The next governance stage can then ask:

What actually occurred during execution, what was observed, what outputs were produced, and can the execution history be reconstructed?

The VEXS internal progression therefore continues:

Execution Basis & Authorization → Execution Instantiation & Control → Execution Observation & Traceability → Deviation & Exception Governance → Execution Completeness & Conformity

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Foundational Principle

Authorized validation becomes governed execution only when the approved validation basis is correctly instantiated, bounded, controlled, and maintained in operational reality.

Canonical Closing Statement

Validation Execution Instantiation & Control Module establishes the operational control layer through which VALIDOS® transforms an authorized Validation Execution Basis into an active and governable Validation Execution Instance. By governing object binding, method instantiation, criteria and evidence activation, source-condition continuity, execution roles, environments, controls, configurations, access, sequencing, dependencies, synchronization, human and automated activity, interventions, overrides, execution states, control loss, recovery, and execution drift, VEICM ensures that validation moves from methodological authorization into operational reality without losing the boundaries, conditions, and controls upon which legitimate Validation Determination ultimately depends.

Parent Resources

→ VALIDOS® Validation Governance Architecture — Validation Governance Layer

→ VALIDOS® Architecture Map

→ VALIDOS® Complete Standards & Modules Index

Internal Module Flow

→ Previous module: Validation Execution Basis & Authorization Module (VEBAM)

→ Next module: Validation Execution Observation & Traceability Module (VEOTM)

Related Documents

→ Validation Execution Standard

→ About Validation Execution Standard

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Canonical Source

https://originopenfoundation.org/